



PROJECT DEFINITION

HIGH-ALTITUDE RESILIENT AI-BASED ANTI-DRONE SYSTEM

Environment-Aware Detection, Precision Tracking & Adaptive Stabilization

PROBLEM STATEMENT ID

SIH1729

THEME

Robotics & Drones / Defence Tech

PROBLEM STATEMENT TITLE

Autonomous Precision Anti-Drone Platform for Extreme High-Altitude Environments

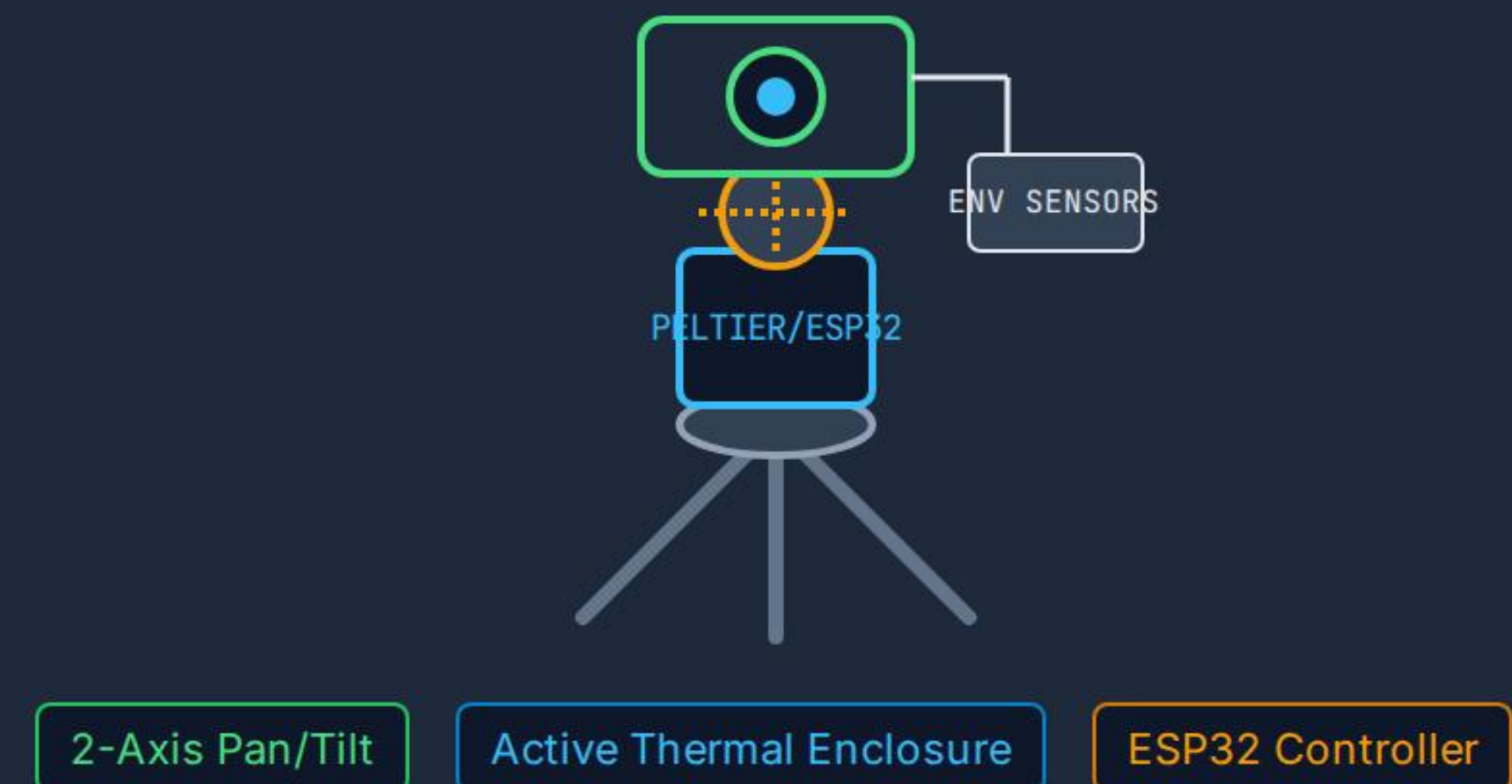
TEAM ID

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TEAM NAME

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SYSTEM HARDWARE CONCEPT DIAGRAM



Core Architecture: A closed-loop, environment-aware anti-drone platform combining AI perception, precision PID gimbal tracking, Peltier thermal management, and continuous ESP32 health monitoring for high-altitude survival.

COMPLETE CLOSED-LOOP CONTROL & COMPENSATION ARCHITECTURE



Closed-Loop AI

Direct coupling of YOLO centroid errors with real-time PID gimbal positioning.

Env Compensation

Environmental disturbances treated as active inputs rather than unknown noise.

Active Thermal Shield

Peltier + heater feedback loop maintaining sensor/electronics baseline.

TCXO/OCXO Monitoring

RF frequency reference tracking to prevent thermal drift during operation.

CURRENT SUBSYSTEM READINESS:

AI DETECTION: IMPLEMENTED

VIDEO INFERENCE: IMPLEMENTED

TRACKING LOGIC: PROTOTYPE

PID GIMBAL: DEVELOPMENT

ENV COMP: DEVELOPMENT

CAD MODEL: DEVELOPMENT

1. AI PERCEPTION LAYER

YOLOv8s Model

60 Epochs | 640×640 Input | Batch Size 16 | MixUp + Copy-Paste Augmentation

Metric (Test: 889 imgs, 906 objs)	Overall	Drone	Bird
Precision	83.5%	92.8%	—
Recall	84.8%	75.8% *	93.8%
mAP@50 / mAP@50-95	81.7% / 51.7%		

* Identified Limitation: Drone recall (75.8%) is the primary bottleneck. Planned next step: dataset expansion for adverse sky conditions.

2. CENTROID TRACKING & PID

Closed-Loop Pan/Tilt



Error Formulation:
$$e_x = x_{\text{drone}} - x_{\text{center}}$$
$$e_y = y_{\text{drone}} - y_{\text{center}}$$

PID updates Pan/Tilt angular velocity to drive $e_x, e_y \rightarrow 0$ continuously.

Encoder Feedback: High-resolution optical encoders provide ground-truth gimbal position feedback to eliminate overshoot.

3. THERMAL & DISTURBANCE COMP

Peltier Loop

- Active Thermal Management:** Peltier element + heater controlled via secondary thermal PID loop inside optical enclosure.
- Mechanical Cable Stiffening:** Temperature sensor inputs adjust PID torque gain to offset cold-weather cable rigidity.
- Wind Loading:** Closed-loop motor current feedback detects wind gust torque disturbances.

Target: Maintain optics & sensors in optimal thermal operating band (-10°C to +40°C) amidst ambient -30°C cold.

4. ESP32 HEALTH & DIAGNOSTICS

Central Microcontroller

ESP32 acts as central hub integrating multi-sensor environmental telemetry:

⚡ Motor Current

Vibration Monitoring

📡 TCXO/OCXO RF Ref

⚠ Real-time Fault Alert

Fail-Safe: Automatic fault detection triggers graceful degradation & backup orientation modes.

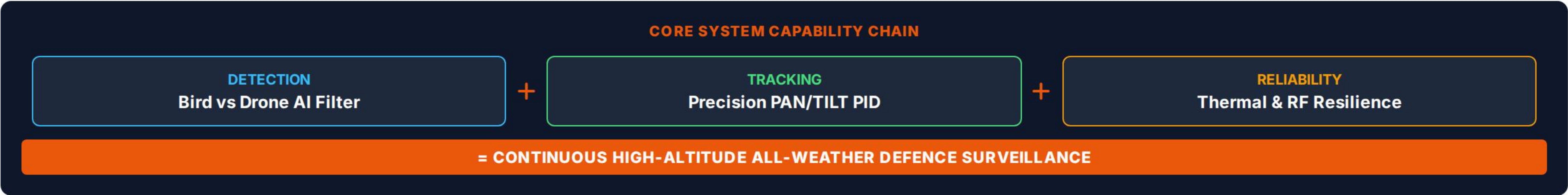
🛡️ Environmental & Hardware Risk Mitigation Matrix

Environmental Risk	Impact on Hardware	Engineering Mitigation Strategy
Extreme Cold (-30°C)	Component degradation, frozen lube	Temp-rated ICs + Active Peltier/Heater closed loop
Cable Rigidity	Gimbal tracking error & resistance	Silicone ultra-flex wiring + temp-based PID offset
High Wind Loads	Pointing disturbance & vibration	Rigid 6061-T6 aluminum structure + closed-loop PID
Sensor / Clock Drift	RF / tracking state instability	TCXO/OCXO frequency ref + continuous zero-point calibration
Drone Misses (Recall 75.8%)	Intermittent target tracking loss	Dataset augmentation + hard-negative mining + temporal filter
Dust & Snow Ingress	Optics blockage & joint wear	IP65 sealed enclosure + heated optical glass window

🔗 Development Roadmap

PHASE 1: AI Object Detection	✓ COMPLETED
PHASE 2: Video Centroid Tracking	✓ PROTOTYPE
PHASE 3: PID Gimbal Integration	→ DEVELOPMENT
PHASE 4: Thermal Compensation Loop	→ DEVELOPMENT
PHASE 5: Hardware System Integration	PLANNED
PHASE 6: High-Altitude Field Test	PLANNED

🔗 Validation Strategy (Strict Protocol):
Simulation → Bench Testing → Controlled Environmental Chamber → High-Altitude Field Validation. *(No unvalidated deployment claimed).*



🛡️ Strategic Asset Protection

- Precision tracking around high-altitude military outposts & borders.
- Eliminates false alarms from high-altitude bird species via AI classification.
- Engagement-interface ready (Detection → Identification → Tracking).

📡 Remote Infrastructure

- Automated monitoring of remote high-altitude energy & telecom sites.
- Reduces reliance on manual observation in sub-zero weather.
- Continuous health reporting minimizes unexpected field maintenance.

💡 Deep-Tech Innovation

- Novel environment-aware control framework combining AI & hardware telemetry.
- Portable 2-axis deployable architecture designed for quick setup.
- Modular platform allowing future sensor payload integration.

Operational Boundary Note: This platform is designed primarily as an **Environment-Aware Detection, Identification & Precision Tracking Platform**. Neutralization interface readiness is provided without claiming unvalidated kinetic/soft-kill mechanisms.

TECHNICAL FOUNDATIONS & LITERATURE

- YOLOv8 Real-time Detection Architecture:** Ultralytics anchor-free object detection framework for small target classification.
- Visual Servo & Centroid Tracking:** Computer vision image-plane error formulation for closed-loop gimbal orientation.
- Closed-Loop PID Control:** Industrial position & velocity control with optical encoder feedback.
- Peltier Thermal Management:** Thermoelectric active heating/cooling loops for sensitive optical electronics.
- TCXO/OCXO Frequency Stability:** Crystal reference monitoring under extreme thermal cycling conditions.
- MIL-STD-810H Standards:** Environmental engineering considerations for high-altitude cold & low pressure.

TEAM EXPERIMENTAL VALIDATION EVIDENCE

YOLOv8s Test Evaluation

Test set: 889 images / 906 objects

Bird Recall: 93.8%
Drone Prec: 92.8%
Drone Rec: 75.8%

Video Inference Pipeline

MP4 / AVI / MOV frame test

best.pt model
Frame bounding box
Centroid extraction

Mechanical Development

2-Axis Pan/Tilt platform

Lightweight portable tripod mount; CAD structural loading analysis for wind/vibration.

Electronics Architecture

ESP32 telemetry hub

Interfacing thermal PID, current sense, vibration & TCXO RF frequency stability.

© Immediate Next Validation Targets:

1. Increase drone recall to >88% | 2. Quantify temperature-induced tracking drift | 3. Characterize wind-load pointing error | 4. Full hardware-in-the-loop closure.

Key Technical Differentiator:

Integrating perception, embedded control, and thermal physics into a unified system rather than treating vision as an isolated software module.